

Daily Logic Drill #7

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Proof by Contradiction · Combining Techniques · Synoptic Mastery.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

The mechanics of proof by contradiction. Pure recognition.

1. To prove statement P by contradiction, you should assume ___ and derive a contradiction.
2. True or false: “Proof by contradiction and proof by contrapositive are the same technique.”
3. What contradiction is reached in the classical proof that $\sqrt{2}$ is irrational?
4. True or false: “If assuming ‘not P ’ leads to a statement that’s always false, P must be true.”
5. Give the first line you would write to prove “there are infinitely many primes” by contradiction.
6. True or false: “Deriving ‘ $1 = 0$ ’ partway through an argument is a valid contradiction to conclude the assumption was false.”
7. True or false: “In a proof by contradiction of ‘if P then Q ’, you assume P is true and Q is false simultaneously.”
8. True or false: “Every valid proof by contrapositive can be rewritten as a proof by contradiction.”
9. To disprove a universal (“for all”) claim as fast as possible, should you generally reach for a counterexample or for proof by contradiction?
10. True or false: “If you can find a single counterexample to a claim, you never need a separate proof by contradiction to disprove that same claim.”

Section B Manipulation Drills

(≤50 s each)

Write short, complete proofs by contradiction.

1. Prove by contradiction: “There is no smallest positive rational number.”
2. Prove by contradiction: “If n^2 is even, then n is even.”
3. Prove by contradiction: “There is no largest even integer.”

4. Prove by contradiction: “If a, b are integers and $a + b$ is odd, then a and b cannot both be even.”
5. Prove by contradiction: “If p is prime and $p > 3$, then p is not divisible by 6.”
6. Prove by contradiction: “There is no integer n satisfying both $n > 5$ and $n < 3$.”
7. Prove by contradiction: “If x is a positive real number, then $x + \frac{1}{x} \geq 2$.”
8. Prove by contradiction: “If n is a positive integer, n and $n + 1$ share no common factor greater than 1.”
9. Prove by contradiction: “ $\log_2 3$ is irrational.”
10. Prove by contradiction: “If a, b are positive integers with $a < b$, then $a^2 < b^2$.”

Section C Substitution & Structure

(≤ 90 s each)

Longer contradiction proofs, combining tools from across the whole pillar.

1. Prove by contradiction: “ $\sqrt{2} + \sqrt{3}$ is irrational.”
2. Prove by contradiction: “There is no integer solution to $x^2 + y^2 = 4z + 3$.”
3. You are given the identity $2^n - 1 = (2^a - 1)(2^{n-a} + 2^{n-2a} + \dots + 1)$ whenever a divides n . Using proof by contradiction, show that if $2^n - 1$ is prime then n is prime.
4. Prove by contradiction: “If n is a positive integer that is not a perfect square, then $\sqrt{n}$ is irrational.”
5. Using proof by contradiction and the fact that 2 is the only even prime (Day 1, B9), prove there do not exist two distinct primes p, q , both greater than 2, with $p + q = 5$.
6. Using contradiction and the fact that $n^2 - n$ is always even (Day 1, D5 / Day 6, C5), prove there is no positive integer n for which $n^2 - n = 7$.
7. Prove by contradiction: “The equation $x^2 + y^2 = 3$ has no solution in positive integers.”
8. Using contradiction, prove that if a, b, c are positive integers with $a^2 + b^2 = c^2$ and c is even, then a and b cannot both be odd.

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Consider: I. “Proof by contradiction and proof by contrapositive always produce identical proofs.” II. “A single counterexample is a special case of proof by contradiction.” III. “If a statement’s negation leads to ‘ $1 = 0$ ’, the statement is proven true.” Which are true?

A) I only

- B) II only
 - C) III only
 - D) I and III only
 - E) I, II and III
2. Prove, using proof by contradiction, that there is no positive integer n for which both n and $n + 1$ are perfect squares.
3. In a room of at least 2 people, every person is either a knight (always tells the truth) or a knave (always lies). Every person in the room says: “Everyone else in this room is a knave.” Using proof by contradiction, determine exactly how many knights are in the room.
4. Using proof by contradiction, prove that the equation $n^2 = 2^k$ has no solution in positive integers n, k with k odd.
5. Give a second, different proof that $\sqrt{2}$ is irrational (Day 5, D2 gave one proof already), this time using the same 2-adic valuation technique as D4.

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths